

HOW SUSTAINABLE IS ETHANOL?

THERE ARE CONCERNS REGARDING ETHANOL PRODUCTION, NOT LEAST AMONG THEM THE WATER INTENSITY OF CROPS SUCH AS PADDY, MAIZE AND SUGARCANE THAT CURRENTLY PREDOMINATE THIS SPACE

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A S INDIA FACES its biggest energy shock in decades, there are renewed calls to increase ethanol blending in petrol to help reduce crude oil imports and save the country some precious foreign exchange.

The Centre has acted swiftly in this regard, raising ethanol blending from a mandatory 20% (E20) to as much as 30% on a non-mandatory basis in the near term and to 85% and 100% in the future. Plus, it has also allowed ethanol blending in aviation fuel.

However, the measures, along with an order regulating the khand-sari (raw cane sugar) industry, which now stands withdrawn, have raised food security fears afresh.

Unlike in 2020, when the government allowed distilleries to use foodgrains like rice, maize and wheat and expanded soft loans to

meet the E20 target, the risks are now tangible.

Industry data on ethanol supply to public sector oil marketing companies (OMCs) show that grains have overtaken traditional sugarcane as feedstock. From zero in Ethanol Supply Year (ESY) of 2017-18, the share of grains jumped to 59.7% in 2022-23 and to 72.4% in ESY 2025-26. Maize and rice now account for 45.2% and 27.2%, respectively, of the total. The ESY runs from November 1 to October 31.

Besides, there are other concerns regarding ethanol production, not least among them the water intensity of crops such as paddy, maize and sugarcane that currently predominate this space.

The industry, however, is ecstatic considering the new measures. The reason? It had sought an increase in blending to 22% to uti-

WHY THE SHIFT

Ethanol blending is benefiting farmers, saving forex and reducing carbon footprint

Higher incentive for ethanol is reshaping cultivation and raising water stress

Foodgrains have overtaken sugarcane as ethanol feedstock, risking food security

Farmers are shifting to maize and away from pulses, oilseeds and other cereals

The focus should now be on 2G and 3G ethanol from agri waste and other non-food items

